SOCNet: A Lightweight and Fine-Grained Object Recognition Network for Satellite On-Orbit Computing

Yanhua Pang, *Graduate Student Member, IEEE*, Yamin Zhang, Yi Wang, Xiaofeng Wei, and Bo Chen, *Member, IEEE*

***Abstract*— High-quality remote sensing images (RSIs) not only provide opportunities for deep-learning-based image interpretation but also challenge satellite edge devices for storage, processing, and downlinking to the ground server. Satellite on-orbit object recognition is an effective measure to solve this challenge. However, there is an insurmountable gap between the extremely limited computing and storage resources of satellites and the demand for those of deep learning. To solve this problem, a fast and lightweight intelligent satellite on-orbit computing network (SOCNet) is proposed. First, the overall network architecture based on the idea of flat multibranch feature extraction is proposed to accelerate model inference and reduce the network depth. Second, we propose the idea of exchanging a larger receptive field for network depth and combine the idea of depthwise separable convolution to further reduce the amount of parameters. We design a feature extraction method of coupled fine–coarse-grained for efficient feature extraction. Finally, global average pooling is used for feature fusion to further reduce the network parameters' amount and computational complexity. We have performed multiple sets of ablation experiments and some state-of-the-art comparison experiments on the server and carried out experimental verification on the NVIDIA Jetson TX2 mobile device to mimic the resource-constrained environment on the satellite. SOCNet with parameters of 0.21 MB, model size of 1.22 MB, and computation cost of 50.81 MB floating point operations (FLOPs) has achieved 99% comprehensive accuracy, 2.45-ms latency, and 408 images per second throughput on 30 types of aircraft datasets with 224 × 224 RSIs.**

***Index Terms*— Fine-grained object recognition, lightweight neural network, remote sensing image (RSI), satellite on-orbit computing.**

I. Introduction

THE application mode of ground processing for acquiring data on traditional satellites is poor in timeliness, the fastest response time is hourly, and it occupies a large amount of satellite storage and satellite-to-ground transmission bandwidth. The high-throughput satellite data volume reaches tens to hundreds of Gbps level, and satellite on-orbit computing is an inevitable choice to improve the timeliness of satellite information application. Satellite on-orbit object recognition is an unmanned edge computing task, so the on-orbit object recognition methods must have fast recognition speed and high recognition accuracy [1]. Due to the limitations of the computing resources, storage resources, heat dissipation, and power consumption, it is difficult to guarantee both the accuracy and speed of on-orbit object recognition, and only a tradeoff between them can be taken. Facing the prominent contradiction between satellite edge equipment with extremely limited resources and high accuracy [2], high timeliness, and high object recognition throughput, it is urgent to study the lightweight, fast, and high-accuracy edge computing network.

Object recognition is a key technical requirement in satellite on-orbit computing tasks. With the rapid development of deep learning [3] and the rapid growth of high temporal and spatial resolution remote sensing images (RSIs) [4], the deep-learning-based object recognition methods have become the focus and hotspot of research [5], [6], [7]. At present, the research on RSI object recognition based on the ground server is quite common, but there are few research on satellite on-orbit object recognition. It is difficult for satellite edge devices with extremely limited computing and storage resources to run deep learning models with huge parameters. Generally, the recognition performance improves with the scale of hyperparameters [8]. So there is a gap between satellite edge equipment with extremely limited resources and high object recognition performance. How to design a lightweight and fast object recognition method to adapt to extremely resource-constrained satellite edge devices is the first problem to be solved in this article.

The application of deep learning on satellite boards is the future development trend of smart satellites. The National Aeronautics and Space Administration (NASA) designed the SpaceCube Edge TPU SmallSat Card for autonomous operations and onboard science data analysis [9]. The European Space Agency (ESA) has developed the first on-board deep neural network demonstrator for satellite Earth observation ϕ-sat-1 [10]. ESA launched OPS-SAT on December 18, 2019, and this nanosatellite pioneering used artificial intelligence (AI) on-orbit applications for autonomous operations. Kacker et al. [11] discussed the deployment of the image processing algorithms developed for BeaverCube-2, a project under development between the MIT Space

Manuscript received 1 September 2022; revised 11 October 2022; accepted 18 October 2022. Date of publication 19 October 2022; date of current version 7 November 2022. This work was supported in part by the Natural Science Foundation of Guangdong Province under Grant 2022A1515010113 and in part by the Shenzhen Natural Science Foundation. *(Corresponding author: Bo Chen.)*

The authors are with the Institute of Space Science and Applied Technology, Harbin Institute of Technology, Shenzhen 518055, China (e-mail: hitchenbo@hit.edu.cn).

Digital Object Identifier 10.1109/TGRS.2022.3216215

1558-0644 © 2022 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 19:48:10 UTC from IEEE Xplore. Restrictions apply.

Telecommunications, Astronomy, Radiation (STAR) Lab and the Northrop Grumman Corporation. The above-mentioned satellite on-orbit computing research has achieved remarkable results. However, research on satellite on-orbit fine-grained object recognition is yet to be carried out. Unlike on the ground where object recognition can be corrected by manual discrimination, the unmanned satellite on-orbit object recognition task needs a high-precision and fine-grained object recognition method. This is another problem that this article aims to solve.

To solve the above problem, we propose a fast and lightweight satellite on-orbit fine-grained object recognition model, satellite on-orbit computing network (SOCNet). Our main contributions are summarized as follows.

1) Given the limited resources on the satellite and the requirements of the satellite on-orbit computing on the response speed and recognition accuracy, an SOCNet based on the idea of flat multibranch feature extraction is proposed.
2) To improve the large-scale feature extraction capability, feature extraction efficiency, and reduce the computational cost, we innovatively propose the idea of exchanging a larger receptive field for network depth and combine the idea of depthwise separable convolution (DSC) to further reduce the amount of parameters. We designed an ultralightweight convolution unit based on them, depthwise separable hybrid dilated convolution (DSHDC).
3) Aiming at the problem of low accuracy of the object recognition methods based on RSIs due to the difficulty in extracting effective features, a feature extraction method of coupled fine–coarse-grained is proposed. Specifically, we innovatively design a coupled fine–coarse-grained module (CFC-module), which is a parallel feature extraction structure to extract features from large-scale and fine-grained, and use the attention layer for feature fusion to enhance key feature expression.

The rest of this article is organized as follows. Section I introduces the related work to SOCNet. Section II details our proposed SOCNet. Section III demonstrates the experimental results on the ground server and the satellite board to evaluate the effectiveness of the proposed approach. Finally, Section IV draws conclusions.

II. Related Work

The storage and computing resources on satellites are limited and precious, so the satellite on-orbit intelligent object recognition model must be a lightweight neural network model to adapt to the resource-constrained on-orbit processing platform. Second, satellites perform on-orbit object recognition tasks, unlike ground platforms that can perform error correction and tune with human intervention, so a fine-grained object recognition model is needed to improve object recognition accuracy. So in this section, we review the related work from three aspects: 1) generic lightweight neural network model; 2) lightweight network for satellite on-orbit computing; and 3) fine-grained object recognition.

A. Generic Lightweight Neural Network Model

Generally, the existing advanced deep learning technologies consume a lot of storage and computing resources, are usually deployed on servers or large cloud computing platforms, and cannot be applied to edge devices with limited resources [12]. The lightweight neural network is the last mile to get through the application of deep learning to edge devices. Iandola et al. [13] proposed SqueezeNet in 2016, which has 50 times fewer parameters than the AlexNet method and almost unchanged accuracy, to reduce the parameters mainly by replacing 3×3 convolution with 1×1 convolution. Howard et al. [14] proposed MobileNet for mobile vision applications in 2017, in which a DSC was designed that could reduce the parameters and computational cost. Based on MobileNet, they proposed MobileNetv2 [15] in 2018 and creatively designed the linear bottleneck structure and the inverse residual structure. Zhang et al. [16] designed ShffleNet, which reduces the parameters and computational cost using group convolution. ShffuleNetv2 [17], MixConv [18], EfficientNet [19], and MobileNetv3 [20] all use neural architecture search (NAS) [21], [22], [23] to optimize accuracy and efficiency. The lightweight neural network proposed in this article is inspired by the MobileNet series methods [14], [15], [20] and the ShffuleNet series methods [16], [17] and draws on dilated convolution [24] which can obtain a large receptive field without increasing the size of the convolution kernel, the parameters of the model, and the computational cost of the model. Lv et al. [25] designed step activation quantization to accelerate the CNN-based RSI object recognition network and achieved a tenfold acceleration effect with slightly lower accuracy.

Most existing lightweight object recognition networks are designed for natural images and rarely for RSIs. Compared with natural images, high-resolution RSIs have changeable directions, complex types and numbers of object, scarcity of samples in specific fields (such as missile positions), and only a single imaging perspective. In addition, different platforms and weather conditions will have an impact on the acquisition of RSIs. These factors lead object recognition in high-resolution RSIs into greater challenges than natural image interpretation. Therefore, the existing lightweight object recognition networks have low recognition accuracy in the RSI datasets.

B. Lightweight Network for Satellite On-Orbit Computing

There are few research on satellite on-orbit object recognition. To adapt to the extremely limited resources of satellite edge devices, Maskey and Cho [26] proposed CubeSatNet, which is an ultralight convolutional neural network designed for on-orbit binary image classification on a 1U CubeSat. But it cannot handle the task of fine-grained object recognition for multiple categories. Qiao et al. [27] proposed LiteSCANet, which is an efficient lightweight network based on spectral and channelwise attention for hyperspectral image classification. Paoletti et al. [28] proposed Ghostnet for hyperspectral image classification. Jia et al. [29] proposed a lightweight CNN for hyperspectral image classification. Cui et al. [30]

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 19:48:10 UTC from IEEE Xplore. Restrictions apply.

proposed LiteDepthwiseNet, which is a lightweight network for hyperspectral image classification. The above lightweight networks are designed for classification of hyperspectral images not involving the research on ordinary RSIs. There are few lightweight networks [31], [32] designed for ordinary RSIs, but they do not fully consider the extremely limited satellite on-orbit computing and storage resources. We design an ultralightweight object recognition network SOCNet for RSIs.

C. Fine-Grained Object Recognition

Fine-grained object recognition is a particularly challenging task, which usually has the characteristics of small interclass variance and large intraclass variance, aiming to distinguish its subcategories from the same category [33]. The fine-grained object recognition methods based on deep learning are generally divided into four categories: 1) general CNN for fine-grained object recognition [34], [35]; 2) part detection and alignment-based approaches [36], [37]; 3) ensemble of network-based approaches [38], [39]; and 4) visual-attention-based approaches [40], [41].

The above methods are suitable for on-ground fine-grained recognition tasks, but those models are developed on ground servers without the burden of computing and storage resources, but they cannot function properly on satellites with severely limited resources. The existing lightweight fine-grained object recognition models for computer vision are developed for natural scenes and are not optimized for RSIs. The recognition performance of them becomes poor when applied directly to the RSI dataset. Natural images are generally well lit, with high color saturation and richer object information. There is a world of difference between RSIs and natural imagery. Due to large strabismus imaging and various disturbances of the imaging environment, the RSI resolution changes unevenly or some pixels are missing, resulting in defective distorted images. Therefore, the object features of RSIs are more difficult to be extracted by models than natural images. In addition, the differences between fine-grained object classes are not obvious, which makes it more difficult for the fine-grained object recognition models based on RSIs to extract object features. So it is necessary to study large-scale and fine-grained feature extraction methods. We propose the coupled fine–coarse-grained feature extraction method and large-scale feature extraction module DSHDC to solve this problem.

III. Methodology

SOCNet is an ultralightweight and fine-grained object recognition network for satellite on-orbit computing. SOCNet is designed to be lightweight from both the overall architecture and the local modules, reducing the depth, parameters' amount, size, and computation of model while taking into account multiscale, coarse-grained, and fine-grained feature extraction. It can be used on resource-constrained satellites to carry out fast and efficient fine-grained object recognition. First, the overall architecture based on the idea of flat multibranch feature extraction adopts multiscale residual parallel structure to enhance the expression of features while reducing the network depth. Second, in the local module, to improve the large-scale feature extraction capability and feature extraction efficiency and reduce the computational cost, we innovatively propose the idea of exchanging a larger receptive field for network depth and design a DSHDC combined with the idea of DSC [42], which greatly reduces the number of parameters and computation. Aiming at the problem of low accuracy of the object recognition methods based on RSIs due to the difficulty in extracting effective features, a feature extraction method of coupled fine–coarse-grained (CFC-module) is proposed. In this method, we innovatively design a parallel feature extraction structure to extract features from two dimensions, large-scale and fine-grained, and use the attention layer for feature fusion to enhance key feature expression. Finally, at the end of the network, the global average pool is used for feature fusion to further reduce the number of network parameters and computational complexity. This model can save limited computing power on satellite and improve the timeliness and accuracy of satellite fine-grained object recognition.

In this section, we will describe in detail the design principle of SOCNet, the formula derivation of related lightweight networks, and the process flow of satellite on-orbit object recognition based on SOCNet, mainly from four aspects to introduce separately: 1) the flat multibranch feature extraction architecture of SOCNet; 2) the idea of exchanging a larger receptive field for network depth; 3) the feature extraction method of coupled fine–coarse-grained (CFC-module); and 4) satellite on-orbit object recognition based on SOCNet.

A. Flat Multibranch Feature Extraction Architecture

According to the idea of lightweight flat efficient feature extraction and the efficient computing principle of shared feature reuse, to avoid repeated extraction of shared features and improve the computing efficiency, the proposed SOCNet is divided into four parallel feature stream branches after the shallow extraction of the Focus module, as shown in Fig. 1.

Assuming that the input RSI size is $M \times N \times 3$, the Focus module first transforms it into $(M/2) \times (N/2) \times 12$ by the slice operation and then obtains the feature map through a layer of 3×3 convolution, as shown in Fig. 2. Let I_{RSI} represent the original input RSI, the Focus module can be formulated as

$$F_{\mathrm{FOCUS}} = F_{\mathrm{focus}}(I_{\mathrm{RSI}}) = \mathrm{Conv2D}(\mathrm{Split}(I_{\mathrm{RSI}})) \tag{1}$$

where $F_{\mathrm{focus}}(\cdot)$ is the feature extraction function of the Focus module and can be divided into the shallow feature extraction step $\mathrm{Conv2D}(\cdot)$ and the slicing operation $\mathrm{Split}(\cdot)$. F_{FOCUS} denotes the output feature map of the Focus module. The Focus module transfers the spatial information of the image into the channel during the slicing operation and achieves two times downsampling without losing information, which reduces the computation caused by the downsampling operation and accelerates the inferencing of the model.

Inspired by residual block feature multiplexing, Branch 1, Branch 2, Branch 3, and Branch 4 are added. But different from the residual block, the four parallel feature stream branches represent four-scale feature extraction:

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 19:48:10 UTC from IEEE Xplore. Restrictions apply.

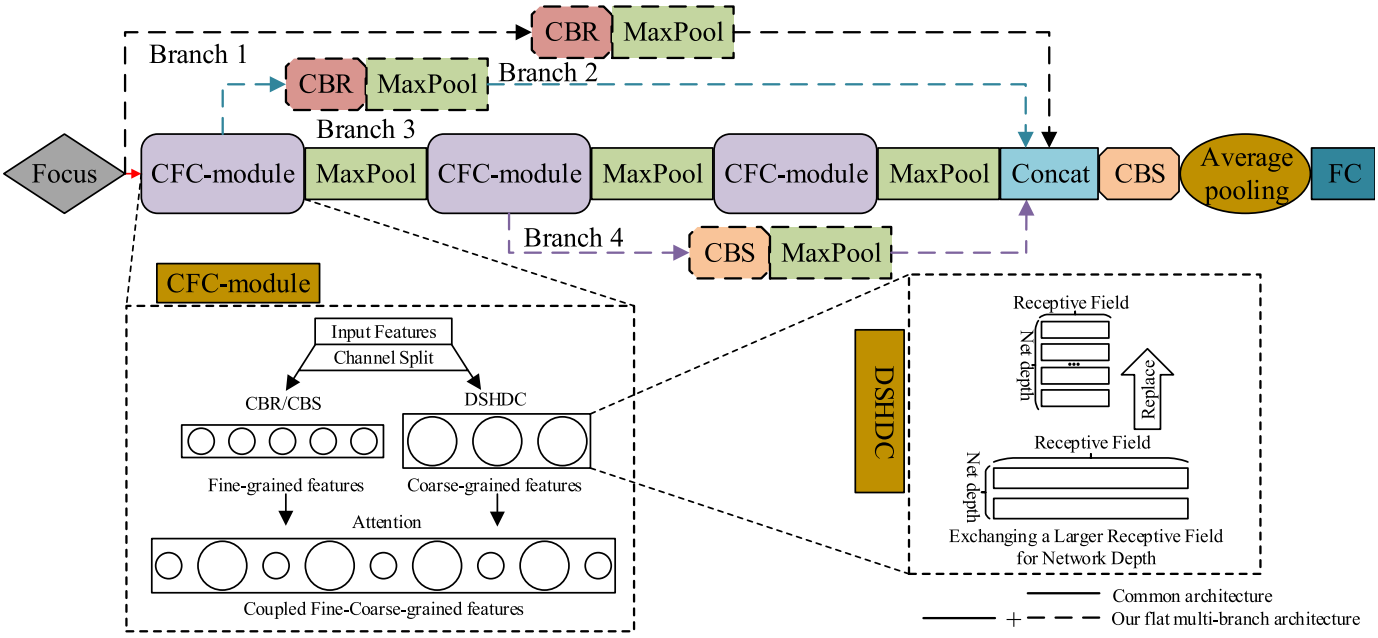


Fig. 1. Framework of the proposed SOCNet.

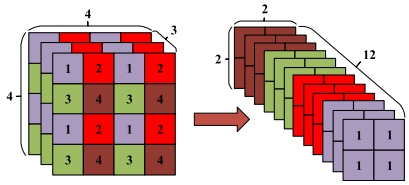


Fig. 2. Schematic of Focus.

Branch 1 represents the original shallow feature extraction, where CBR represents Conv2D, BatchNorm, and ReLU operation; Branch 3 represents Scale 3 feature extraction because this branch contains three CFC-modules; Branch 2 and Branch 4 represent Scale 2 feature extraction and Scale 1 feature extraction, respectively, where CBS represents Conv2D, BatchNorm, and SiLU operation. It is acknowledged that ReLU is better than the SiLU function in the shallow feature extraction of the network. At the end of SOCNet, CBS and global average pooling (GAP) are used for feature fusion, and the fused features are used as the input of the fully connected (FC) layer with softmax activation function, and finally, the recognition result is obtained. The detailed configuration of the network parameters of SOCNet is shown in Table I.

The framework of SOCNet has four branches instead of more or less, as well as its network depth, which is the result of tradeoffs between the number of parameters, computation, and identification accuracy.

TABLE I
SOCNET NETWORK PARAMETER CONFIGURATION

Branch	Layer	Kernel	Strides	Activation	Padding
1234	Focus	1	1	ReLu	None
1	CBR	1	1	ReLu	None
1	MaxPool	3	8	-	-
2	CFC-module	-	-	-	-
4	CBR	1	1	ReLu	None
4	MaxPool	3	8	-	-
2	MaxPool	3	2	-	-
2	CFC-module	-	-	-	-
3	CBS	1	1	SiLu	None
3	MaxPool	3	4	-	-
2	MaxPool	3	2	-	-
2	CFC-module	-	-	-	-
2	MaxPool	3	2	-	1
1234	concat	-	-	-	-
1234	CBS	1	1	SiLu	None
1234	GAP	1	-	-	-
1234	FC	23	-	-	-

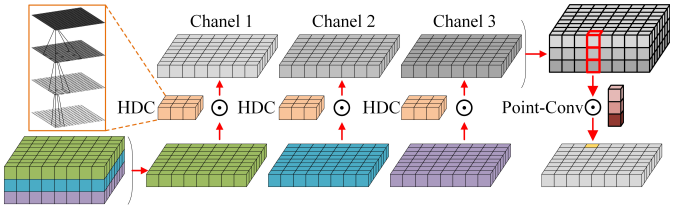


Fig. 3. Schematic of DSHDC.

B. Exchanging a Larger Receptive Field for Network Depth

To adapt to the limited resources on the satellite, reduce the parameters and computational cost, and improve the recognition efficiency, according to the idea of exchanging a larger receptive field for network depth and combining with the idea of DSC, we designed an ultralightweight convolution unit, DSHDC, as shown in Fig. 3. DSHDC replaces depth convolution with hybrid dilated convolution (HDC) to obtain a larger receptive field and improve the efficiency of object feature extraction.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 19:48:10 UTC from IEEE Xplore. Restrictions apply.

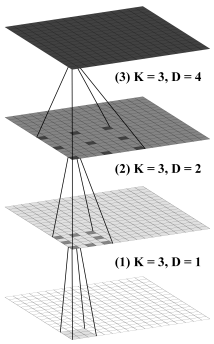


Fig. 4. How HDC efficiently extracts large-scale receptive field. (1) F_1 is produced from F_0 by a 1-dilated convolution; each element in F_1 has a receptive field of 3×3. (2) F_2 is produced from F_1 by a 2-dilated convolution; each element in F_2 has a receptive field of 7×7. (3) F_3 is produced from F_2 by a 4-dilated convolution; each element in F_3 has a receptive field of 15×15.

HDC [24] is an efficient convolution operation, which can achieve a larger scale receptive field without increasing the parameters and computational cost. Fig. 4 shows how HDC efficiently extracts the large-scale receptive field.

We define F_i to be the output feature map of the ith layer. It is easy to see that the size of the feature map of each element in F_i is

$$F_{i+1} = \left(2^{i+2} - 1\right)^2 \quad \text{for} \quad i = 1, 2, \ldots, n-2. \tag{2}$$

The feature map is a square of exponentially increasing size. $F_3 = 15$ means that HDC only uses three layers of 3×3 dilated convolution to obtain a receptive field of 15×15. Also, to obtain a 15×15 receptive field, standard convolution needs to use a 15×15 convolution kernel, assume that the dimensions of the input feature graph and the output feature graph are C_1 and C_2, respectively, and the ratio of the parameters of the former P_{HDC} to the latter P_{standard} is

$$\frac{P_{\text{HDC}}}{P_{\text{standard}}} = \frac{3 \times C_1 \times C_2 \times 3 \times 3}{C_1 \times C_2 \times 15 \times 15} = \frac{3}{25} = 0.12. \tag{3}$$

That is, the HDC uses 12% of the parameters of the standard convolution to obtain the same receptive field as the standard convolution.

HDC is a stack of three layers of 3×3 convolutions with different dilated rates, that is, DSHDC performs a pointwise convolution after three depthwise convolutions. The parameters of DSHDC are $3 \times 3 \times 3 \times C_1 + C_1 \times C_2$, the parameters of three DSC are $3 \times (3 \times 3 \times C_1 + C_1 \times C_2)$, and the ratio of the parameters of the former P_{DSHDC} to the latter P_{DSC} is

$$\frac{P_{\text{DSHDC}}}{P_{\text{DSC}}} = \frac{3^3 \times C_1 + C_1 \times C_2}{3 \times \left(3^2 \times C_1 + C_1 \times C_2\right)} = \frac{9 + \frac{C_2}{3}}{9 + C_2} < 1. \tag{4}$$

Therefore, the DSHDC proposed in this article is lighter than the depth separable convolution, especially when the output feature map size C_2 is larger.

This unit draws on the advantages of fewer parameters and less computational cost of DSC [43] and the advantages of large-scale feature extraction of HDC. Based on ensuring feature extraction capabilities, it greatly reduces the network parameters, network depth, computational cost, and model size.

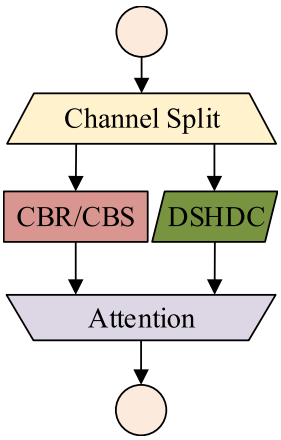


Fig. 5. Schematic of the CFC-module.

TABLE II
DETAILED NETWORK PARAMETER CONFIGURATION OF CFC-MODULE

Branch	Layer		Kernel	Strides	Activation	Padding	Dilation
	Channel Split		-	-	-	-	-
Left	CBR/CBS		1	1	ReLu/SiLu	0	1
Right	DSHDC	HDC	3	1	ReLu/SiLu	1	1
		HDC	3	1	ReLu/SiLu	2	2
		HDC	3	1	ReLu/SiLu	5	5
		PointConv	1	1	ReLu/SiLu	0	1
	Attention	SElayer	-	-	-	-	-

C. Coupled Fine–Coarse-Grained Feature Extraction

The CFC-module is an important multiscale feature extraction module in SOCNet. As shown in Fig. 5, inspired by ShffuleNetv2, the first channel of the CFC-module splits the input features into two parallel branches according to the ratio of 1:1, and the left branch uses CBR/CBS operations, CBR for shallow feature extraction is better than CBS, to extract features for local details. The right branch uses DSHDC operations for large-scale feature extraction, and the output feature maps of the two branches are feature fused by the attention layer [44] to obtain the output feature maps of the CFC-module. Therefore, the CFC-module is capable of both fine-grained extraction of local detail features and coarse-grained extraction of large-scale abstract features. The detailed network parameter configuration of the CFC-module is shown in Table II.

D. Satellite On-Orbit Object Recognition Based on SOCNet

The implementation process of the satellite on-orbit object recognition technology is shown in Fig. 6, which is divided into three stages [10], and the corresponding algorithm flowchart is shown in Fig. 7.

1) Training on the ground server. In the first stage, we construct the historical RSI dataset and use them to train and tune SOCNet on the ground server, and then obtain the trained model to prepare for the second stage.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 19:48:10 UTC from IEEE Xplore. Restrictions apply.

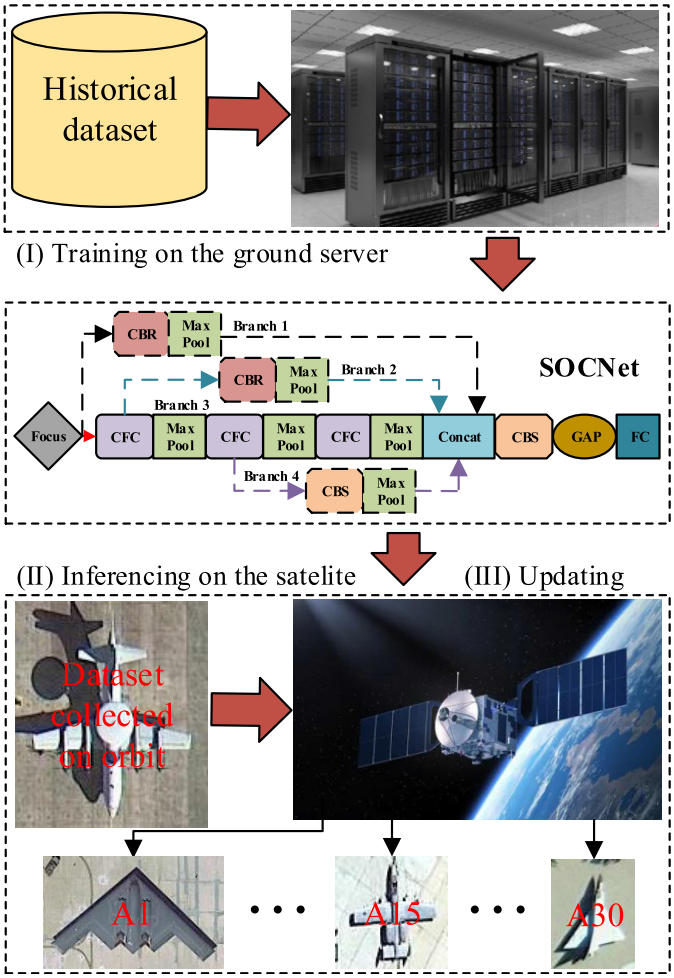


Fig. 6. Satellite on-orbit object recognition based on SOCNet.

The algorithm corresponding to this stage is the blue line in Fig. 7.

2) Inferencing in the satellite. In the second stage, we use the trained model obtained in stage one and deploy it on the satellite board. The RSIs obtained on-orbit will be directly used for object recognition without being transmitted to the ground. The algorithm corresponding to this stage is the red line in Fig. 7.
3) Updating the better model on the satellite. In the third stage, we regularly update the latest optimal model on the ground to the satellite. The better model obtained by fine-tuning the previously trained model using the newly acquired RSI data of the new object is relative to the previously trained model. So the on-orbit model can identify new objects as early as possible. The algorithm corresponding to this stage is the cyan line in Fig. 7.

This method of satellite on-orbit object recognition has changed the traditional RSI satellite-to-ground transmission and ground processing application mode, greatly improving the timeliness of satellite response.

IV. Experimental Settings

In this section, the proposed SOCNet is verified on the ground server and the satellite imitated by TX2, and the results

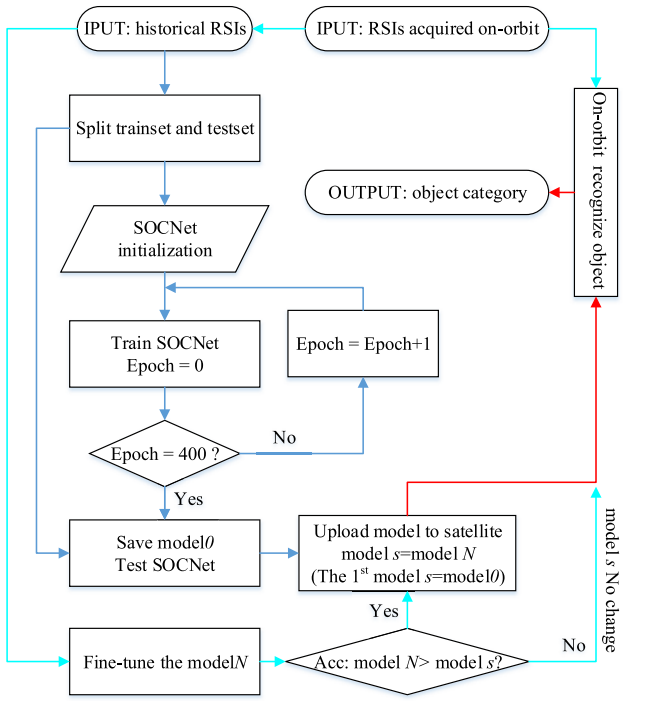


Fig. 7. Algorithm flowchart of satellite on-orbit object recognition based on SOCNet, where model*0* represents the model obtained after the first ground training, model *s* represents the model currently running on the satellite, and model *N* represents the model fine-tuned by the latest acquired data.

of the experiment, including the ablation study and some comparison experiments of the state-of-the-art lightweight methods, will be presented and analyzed.

A. Data Description

There is currently no public RSI dataset for fine-grained recognition tasks, so we have marked a dataset that contains 30 types of aircraft on the WorldView-3 satellite images with a resolution of 0.31 m ourselves, which is called the Aircraft dataset. Detailed information about the Aircraft dataset is shown in Fig. 8; the aircraft dataset has a total of 4378 samples, the size of each sample is $3 \times 224 \times 224$, and on average there should be 146 for each type of aircraft, but we found that the number of A26, A4, and A29 is very small, only 11, 6, and 4, respectively, while the number of A19 is 701, so the dataset has serious data imbalance problem and small samples, and the imbalance ratio (IR) has reached $701{:}4 = 175.25{:}1$. To solve the problem of insufficient samples and unbalanced data of some aircraft categories of the training dataset, we expanded the samples by performing geometric transformation operations in the training dataset, including random horizontal flipping, 30° random rotation, random resize clipping, 30° random affine transformation, and scaling of 1/0.877. However, the test set is kept in its original state, without data enhancement, to reflect the performance of the model under real data conditions. The training set and the test set are divided according to the ratio of 3:1. There are 3324 samples in the training set (including the validation set, which is 1/3 of the training set) and 1114 samples in the test set.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 19:48:10 UTC from IEEE Xplore. Restrictions apply.

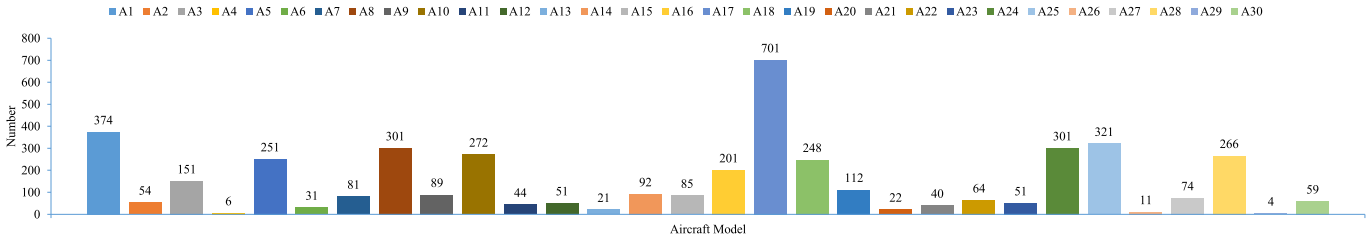


Fig. 8. Aircraft dataset details.

B. Performance Metric

In this article, the fine-grained recognition task is a multiclass classification problem. To evaluate the model performance with unbalanced data, we adopt macro-average F1-score (MacroF 1), macro-average Precision (Macro P), and macro-average Recall (Macro R) as performance metrics. Macro F 1 is a comprehensive metric to evaluate the overall performance in all the classes. F1-score is a comprehensive performance metric for each subcategory. Accuracy is the overall accuracy of all the categories. The specific calculation formula is as follows:

$$\text{Precision}(P) = \frac{t_p}{t_p + f_p} \tag{5}$$

$$\text{Recall}(R) = \frac{t_p}{t_p + f_n} \tag{6}$$

$$\text{f1-score} = \frac{2 \times P \times R}{P + R} \tag{7}$$

$$\text{Accuracy} = \frac{\sum_{i=1}^{x} t_{pi} + t_{ni}}{\sum_{i=1}^{x} t_{pi} + t_{ni} + f_{pi} + f_{ni}} \tag{8}$$

$$\text{Macro } P = \frac{1}{x} \sum_{i=1}^{x} P_i \tag{9}$$

$$\text{Macro } R = \frac{1}{x} \sum_{i=1}^{x} R_i \tag{10}$$

$$\text{Macro F 1} = \frac{2 \times \text{ Macro } P \times \text{ Macro } R}{\text{Macro } P + \text{ Macro } R} \tag{11}$$

where t_p, t_n, f_p, and f_n refer to the true positives, true negatives, false positives, and false negatives in class i, respectively. Herein, x is the number of classes to be classified.

C. Training Details

All experimental model training is carried out on the ground server. The server configuration is as follows: 2 × 10-core Intel Xeon(R) E5-2640v4 at 2.40 GHz CPU, an NVIDIA Corporation TU102 [GeForce RTX 2080 Ti] GPU, an operating system of 64-bit Ubuntu 20.04.1 LTS, the programming language is Python-3.7.4, and the deep learning framework is PyTorch-1.9.0.

The hyperparameters of the model are obtained by a grid search with the following range: batch size: [16,32,64,128], epochs: [100,200,300,400,500], optimizer: [adam, Adadelta, stochastic gradient descent (SGD)], and we establish a five-fold cross-validation. The model hyperparameters are set as follows: the optimizer is SGD (momentum = 0.9), the initial learning rate is 0.02, the attenuation is 0.1 at 200 and 300 epochs, the batch_size is 64, the epoch is 400, the loss function is cross-entropy, and the random seed is 42.

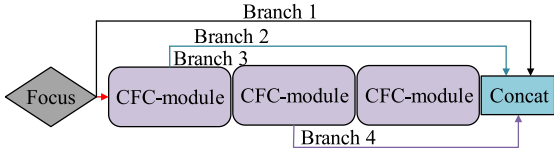


Fig. 9. Schematic of the SOCNet architecture.

The method of model hyperparameter fine-tuning: Do not freeze any layers of the model, use a large learning rate (0.02) for the last FC layer, and use a small learning rate (0.00001) for other layers of the model. The full model is trained for 200 epochs to obtain the fine-tuned model.

D. Ablation Study

To highlight the contribution of SOCNet proposed in this article in the lightweight model architecture, ultralightweight computing unit, and multiscale feature extraction module, as well as the advantages of such design, we set up related ablation experiments.

1) Study of the Flat Multibranch Feature Extraction Architecture. Discuss the advantages of the multibranch lightweight architecture with feature reuse compared with the common multibranch architecture and explore the branch number (network width) and network depth of SOCNet.

2) Study of the Ultralightweight Convolution Unit. Discuss the advantages of using an ultralightweight convolution unit DSHDC compared with DSC and HDC.

3) Study of the Coupled Fine–Coarse-Grained Feature Extraction. Discuss the advantages of using the multiscale efficient feature extraction module CFC-module compared with only using DSHDC.

1) Study of the Flat Multibranch Feature Extraction Architecture: The features of the four branches of SOCNet in Fig. 9 are multiplexed. For example, the feature map output by the first CFC-module is used by Branch 2, Branch 3 and Branch 4, and the feature map output by the second CFC-module is used by Branch 2 and Branch 3. Compared to the SOCNet structure, SOCNetA does not use feature multiplexing, as shown in Fig. 10, four branches are still used. Branch 1 and branch 2 are the same as SOCNet. Branch 3 and branch 4, respectively, use two CFC-modules and one CFC-module independently. And the other network details in Fig. 10 are the same as those in Fig. 9. SOCNetB is the result

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 19:48:10 UTC from IEEE Xplore. Restrictions apply.

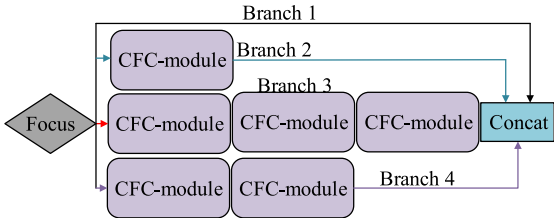


Fig. 10. Architecture of SOCNet$_A$ without feature multiplexing.

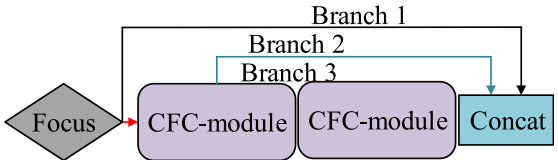


Fig. 11. Architecture of SOCNet$_B$ reduces the width and depth of the model.

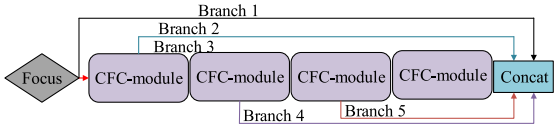


Fig. 12. Architecture of SOCNet$_C$ increases the width and depth of the model.

of SOCNet removing Branch 4 and reducing a CFC-module. Although this reduces the width and depth of the model, as shown in Fig. 11, its accuracy is reduced by 18% compared to SOCNet. SOCNetC adds Branch 5 and a CFC-module on the basis of SOCNet, as shown in Fig. 12. This increases the width and depth of the model. Compared with SOCNetB, its accuracy is increased, but the model's inference speed is greatly reduced.

From the experimental results in the Tab. III, Fig. 13 and Fig. 14, it can be seen that the structure of SOCNet is better than SOCNetA in terms of model size, computational cost [floating point operations (FLOPs)], parameter amount, throughput (images per second, IPS), recognition accuracy, latency and other comprehensive evaluation indicators. This shows that the structure of SOCNet can reduce the model parameters and size, reduce the computational cost, and improve the accuracy and speed of model inference. From the experimental results in the Tab. III, Fig. 13 and Fig. 14, it also can be seen that the selection of the width and depth of the SOCNet network is reasonable and has excellent performance.

2) Study of Depthwise Separable HDC: The basic component unit of SOCNet is the ultralightweight convolution unit, DSHDC. To prove that the computing unit is lightweight and efficient, we used DSC and HDC to replace the DSHDC of SOCNet, respectively, called SOCNet$_{\text{DSC}}$ and SOCNet$_{\text{HDC}}$, for experiments. It can be seen from the experimental results Table III that SOCNet using DSHDC has a smaller model size and computational cost, and all the comprehensive evaluation indicators of that are higher than those of SOCNet$_{\text{DSC}}$ and SOCNet$_{\text{HDC}}$. Although the parameters of SOCNet$_{\text{DSC}}$ are

TABLE III
EXPERIMENTS RESULTS ON THE GROUND SERVER

Model	Model_size(MB)	FLOPs(M)	Parameters(M)	Macro$P$	Macro$R$	Macro $F1$
SOCNet$_A$	1.46	67.06	0.26	0.89	0.88	0.87
SOCNet$_B$	1.32	52.84	0.15	0.81	0.80	0.79
SOCNet$_C$	1.73	75.28	0.32	0.87	0.86	0.87
SOCNet$_{DSC}$	1.30	255.85	0.14	0.95	0.90	0.92
SOCNet$_{HDC}$	1.71	311.07	0.22	0.90	0.86	0.88
SOCNet$_{DSHDC}$	2.00	64.75	0.41	0.94	0.91	0.92
MobieNetv3small	10.40	58.04	1.26	0.86	0.83	0.84
ShffuleNetv2	10.74	147.00	1.31	0.90	0.85	0.87
MnasNet	25.43	313.13	3.14	0.92	0.92	0.92
SqueezeNet	6.04	737.08	0.75	0.90	0.85	0.87
EMNet	10.58	337.53	2.39	0.93	0.93	0.92
LGRIN	12.32	493.33	4.63	0.55	0.50	0.50
SOCNet	**1.22**	**50.81**	**0.21**	**0.98**	**0.98**	**0.98**

0.07 m less than those of SOCNet, its computational cost is five times that of SOCNet, because SOCNet$_{\text{DSC}}$ uses a lot of 1×1 convolution. The parameters of SOCNet$_{\text{HDC}}$ are 0.01 M more than those of SOCNet, and its computational cost is six times that of SOCNet. According to the results of the inferencing experiment on TX2 in Figs. 13 and 14, the throughput of SOCNet is almost twice that of SOCNet$_{\text{DSC}}$ and SOCNet$_{\text{HDC}}$, the latency of SOCNet is 2.45 ms, and the inference time of SOCNet is 2.73 s, both of which are lower than those of other models, and its recognition accuracy is also higher. The experimental results show that DSHDC reduces the model size and computational cost, increases throughput and the recognition accuracy, and accelerates model inference.

3) Study of Coupled Fine–Coarse-Grained Feature Extraction: The CFC-module is the core multiscale feature extraction module of SOCNet, and it is the key to improving the feature expression ability of the model. We use DSHDC instead of the CFC-module of SOCNet, called SOCNet$_{\text{DSHDC}}$, to perform experiments. From the experimental results in Table III, the model size, computational cost, and parameters of the SOCNet model are less than those of SOCNet$_{\text{DSHDC}}$, and its comprehensive indicators are better than those of SOCNet$_{\text{DSHDC}}$. According to the results of the inferencing experiment on TX2 in Figs. 13 and 14, both the throughput and recognition accuracy of SOCNet are higher than SOCNet$_{\text{DSHDC}}$. The experimental results show that the CFC-module contributes to the model feature expression ability.

E. Comparison With State-of-the-Art Methods

To further prove the advantages of the SOCNet method in lightweight and effectiveness, we use the existing advanced lightweight networks MobiNetv3small [20], ShuffleNetv2 [17], MnasNet [22], SqueezeNet [13], ESPA-MSDWNet (EMNet) [45], and LGRIN [32] to carry out comparative experiments on the ground server and the satellite board.

1) Verification on the Ground Server: From the experimental results in Table III, the model size of SOCNet is about one-tenth of MobiNetv3small, ShuffleNetv2, EMNet, and LGRIN. The computational cost of SOCNet is almost the same as that of MobiNetv3small, about one-third of ShuffleNetv2 and about one-tenth of LGRIN. The parameter amount of SOCNet is almost one-sixth that of MobiNetv3small and ShuffleNetv2, 4.5% of LGRIN and 8.8% of EMNet. Almost all the comprehensive evaluation indicators of SOCNet are 10%

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 19:48:10 UTC from IEEE Xplore. Restrictions apply.

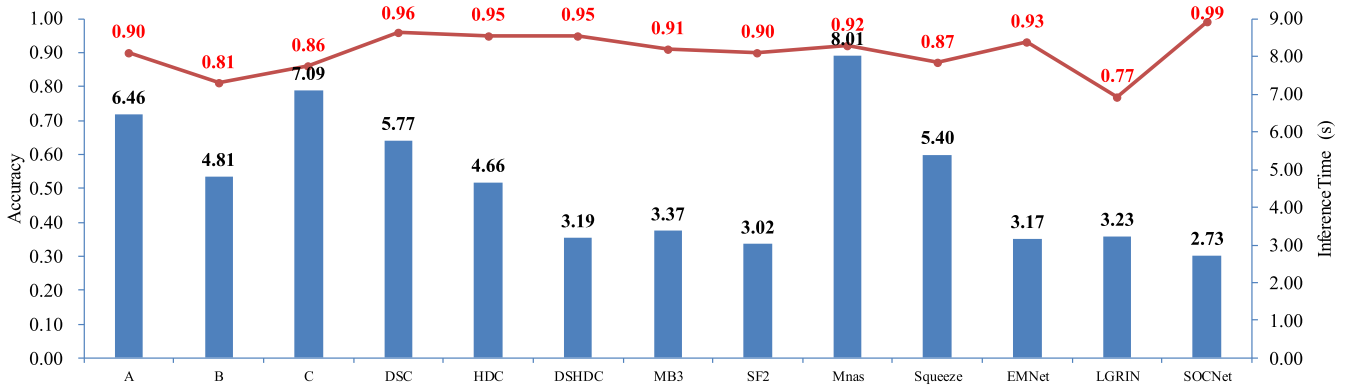


Fig. 13. Inferencing time and recognition accuracy of the SOCNet model on NVIDIA Jetson TX2. The red line is the average precision of fivefold cross-validation, and the blue histogram is the inferencing time of the test dataset with 1114 images. A: SOCNet$_A$; B: SOCNet$_B$; C: SOCNet$_C$; DSC: SOCNet$_{\text{DSC}}$; HDC: SOCNet$_{\text{HDC}}$; DSHDC: SOCNet$_{\text{DSHDC}}$; MB3: MobileNetv3small; SF2: ShffuleNetv2; Mnas: MnasNet; Squeeze: SqueezeNet.

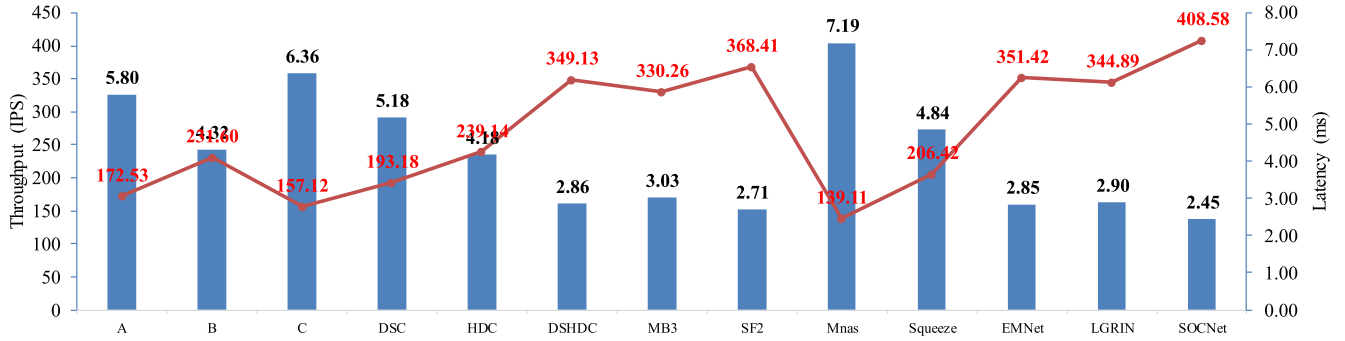


Fig. 14. Throughput and latency of the SOCNet model on NVIDIA Jetson TX2. The red line is the throughput of the inferencing experiment with batch size 128, and the blue histogram is the inferencing Latency. A: SOCNet$_A$; B: SOCNet$_B$; C: SOCNet$_C$; DSC: SOCNet$_{\text{DSC}}$; HDC: SOCNet$_{\text{HDC}}$; DSHDC: SOCNet$_{\text{DSHDC}}$; MB3: MobileNetv3small; SF2: ShffuleNetv2; Mnas: MnasNet; Squeeze: SqueezeNet.

TABLE IV
DETAILED EXPERIMENTAL RESULTS OF ALL THE MODELS ON THE SERVER

	Aircraft	A1	A2	A3	A4	A5	A6	A7	A8	A9	A10	A11	A12	A13	A14	A15	A16	A17	A18	A19	A20	A21	A22	A23	A24	A25	A26	A27	A28	A29	A30
	P	0.95	0.87	1.00	1.00	0.94	0.86	0.88	0.76	0.83	1.00	1.00	0.81	0.91	0.93	0.94	0.88	0.92	0.67	0.79	0.50	1.00	0.87	0.94	1.00	0.84	0.97	0.93	0.00	0.88	1.00
MB3	R	0.94	1.00	0.97	1.00	0.97	0.91	1.00	0.80	0.71	1.00	0.92	0.81	0.95	0.84	0.93	0.92	0.86	0.80	0.65	0.50	0.67	0.89	0.99	0.50	0.89	0.95	1.00	0.00	0.88	0.60
	f1-score	0.94	0.93	0.99	1.00	0.95	0.88	0.94	0.78	0.77	1.00	0.96	0.81	0.93	0.88	0.93	0.90	0.89	0.73	0.71	0.50	0.80	0.88	0.96	0.67	0.86	0.96	0.97	0.00	0.88	0.75
	P	0.96	0.87	0.97	1.00	0.87	0.86	0.96	0.58	0.67	1.00	0.90	0.85	0.95	0.93	0.94	0.94	0.96	0.80	0.71	0.80	0.88	0.85	0.94	1.00	0.94	0.94	0.93	1.00	0.89	1.00
SF2	R	0.95	1.00	0.97	1.00	0.97	0.85	1.00	0.70	0.86	1.00	0.75	0.69	0.95	0.84	0.94	0.97	0.93	0.80	0.87	0.40	0.58	0.88	0.99	0.50	0.83	0.92	1.00	1.00	0.83	0.60
	f1-score	0.95	0.93	0.97	1.00	0.92	0.86	0.98	0.64	0.75	1.00	0.82	0.76	0.95	0.88	0.94	0.95	0.95	0.80	0.78	0.53	0.70	0.86	0.96	0.67	0.88	0.93	0.97	1.00	0.88	0.75
	P	0.97	0.86	0.97	1.00	0.86	0.94	0.96	0.67	0.83	0.92	1.00	0.82	0.91	0.93	0.95	0.92	0.96	0.80	0.85	0.78	0.71	0.90	0.96	1.00	0.88	0.95	0.93	1.00	0.91	1.00
Mnas	R	0.99	0.92	1.00	1.00	1.00	0.91	1.00	0.80	0.71	1.00	0.75	0.88	0.95	0.86	0.95	0.97	0.89	0.80	0.74	0.70	0.42	0.87	1.00	0.50	0.83	0.94	1.00	1.00	0.93	0.60
	f1-score	0.98	0.89	0.99	1.00	0.93	0.93	0.98	0.73	0.77	0.96	0.86	0.85	0.93	0.90	0.95	0.94	0.93	0.80	0.79	0.74	0.53	0.88	0.98	0.67	0.86	0.95	0.97	1.00	0.92	0.75
	P	0.96	0.80	1.00	1.00	0.87	0.86	1.00	0.67	0.83	1.00	0.78	0.71	0.87	0.88	0.89	0.87	0.93	0.80	0.70	0.60	0.62	0.81	0.92	0.50	0.73	0.94	0.87	0.50	0.90	0.60
Squeeze	R	0.92	0.92	0.92	1.00	0.97	0.91	1.00	0.70	0.71	1.00	0.58	0.62	0.95	0.74	0.92	0.89	0.89	0.80	0.61	0.30	0.42	0.89	0.99	0.50	0.61	0.92	0.93	1.00	0.88	0.60
	f1-score	0.94	0.86	0.96	1.00	0.92	0.88	1.00	0.68	0.77	1.00	0.67	0.67	0.91	0.80	0.91	0.88	0.91	0.80	0.65	0.40	0.50	0.85	0.95	0.50	0.67	0.93	0.90	0.67	0.89	0.60
	P	0.94	0.92	0.97	1.00	0.98	0.96	0.88	1.00	1.00	1.00	0.73	0.84	1.00	0.96	0.97	0.78	1.00	0.83	1.00	0.89	1.00	0.95	0.93	1.00	0.90	0.90	0.92	1.00	1.00	0.62
EMNet	R	0.98	0.85	0.95	1.00	0.97	0.95	1.00	0.75	1.00	1.00	0.92	1.00	0.86	0.92	0.86	1.00	1.00	1.00	0.87	0.80	0.58	1.00	1.00	1.00	1.00	0.97	0.86	1.00	0.88	1.00
	f1-score	0.96	0.88	0.96	1.00	0.98	0.95	0.94	0.86	1.00	1.00	0.81	0.91	0.92	0.94	0.92	0.87	1.00	0.91	0.93	0.84	0.74	0.97	0.96	1.00	0.95	0.93	0.89	1.00	0.94	0.77
	P	0.88	0.00	0.82	0.00	0.67	0.64	0.93	1.00	1.00	0.45	0.00	0.00	0.72	0.71	0.84	0.77	0.75	0.00	0.68	0.00	0.17	0.49	0.90	0.00	0.71	0.97	0.67	0.00	0.81	1.00
LGRIN	R	0.90	0.00	0.89	0.00	0.97	0.77	0.64	0.30	0.43	0.91	0.00	0.00	0.62	0.80	0.93	0.79	0.64	0.00	0.83	0.00	0.08	0.52	0.97	0.00	0.28	0.97	0.29	0.00	1.00	0.40
	f1-score	0.89	10.00	0.86	0.00	0.79	0.70	0.76	0.46	0.60	0.61	0.00	0.00	0.67	0.75	0.88	0.78	0.69	0.00	0.75	0.00	0.11	0.50	0.93	0.00	0.40	0.97	0.40	0.00	0.89	0.57
	P	1.00	0.93	1.00	1.00	1.00	1.00	1.00	1.00	0.88	1.00	1.00	1.00	1.00	0.98	0.99	0.98	1.00	0.83	1.00	1.00	0.92	1.00	1.00	1.00	0.95	1.00	1.00	1.00	0.97	1.00
SOCNet	R	1.00	1.00	1.00	1.00	1.00	0.99	1.00	0.95	1.00	1.00	1.00	1.00	1.00	0.98	0.99	1.00	0.96	1.00	1.00	0.90	1.00	1.00	1.00	1.00	1.00	0.98	1.00	1.00	0.99	0.80
	f1-score	1.00	0.96	1.00	1.00	1.00	0.99	1.00	0.97	0.93	1.00	1.00	1.00	1.00	0.98	0.99	0.99	0.98	0.91	1.00	0.95	0.96	1.00	1.00	1.00	0.97	0.99	1.00	1.00	0.98	0.89

higher than those of MobiNetv3small and ShuffleNetv2. MnasNet and SqueezeNet are more expensive than SOCNet in three aspects: parameters' amount, model size, and computation cost, and MnasNet and SqueezeNet have worse recognition performance than SOCNet. The detailed results of the aircraft fine-grained object recognition of all the models are shown in the confusion matrix Figs. 15 and 16. All the elements are gathered on the diagonal, indicating that the recognition accuracy is great. It can be seen from the confusion matrix that SOCNet has the most elements on the diagonal, which shows that SOCNet has the highest recognition accuracy. Figs. 15 and 16 show that the state-of-the-art methods generally perform poorly on the recognition tasks of aircraft-A20 and aircraft-A21, while SOCNet's identification performance is 90% and 100%, respectively, indicating SOCNet's superior performance on the more indistinguishable aircraft types. The detailed evaluation indicators P, R, and F1-score of each aircraft type of all the models are shown in Table IV. It can be seen from the table that many indicators are 1 or close to 1, indicating that SOCNet has excellent recognition performance.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 19:48:10 UTC from IEEE Xplore. Restrictions apply.

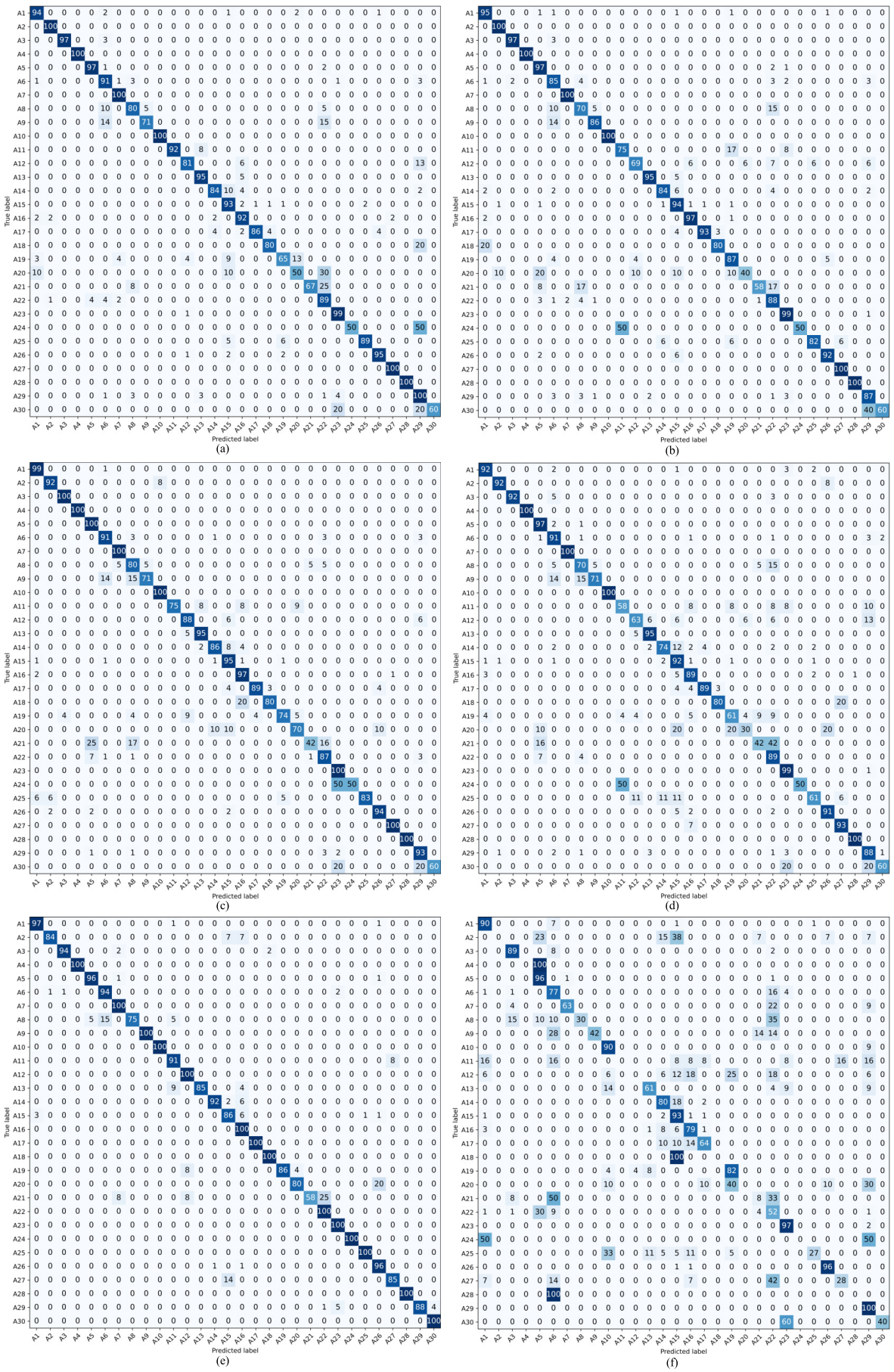


Fig. 15. Normalized confusion matrices of: (a) MobileNetV3small, (b) ShuffleNetV2, (c) MnasNet, (d) SqueezeNet, (e) EMNet, and (f) LGRIN.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 19:48:10 UTC from IEEE Xplore. Restrictions apply.

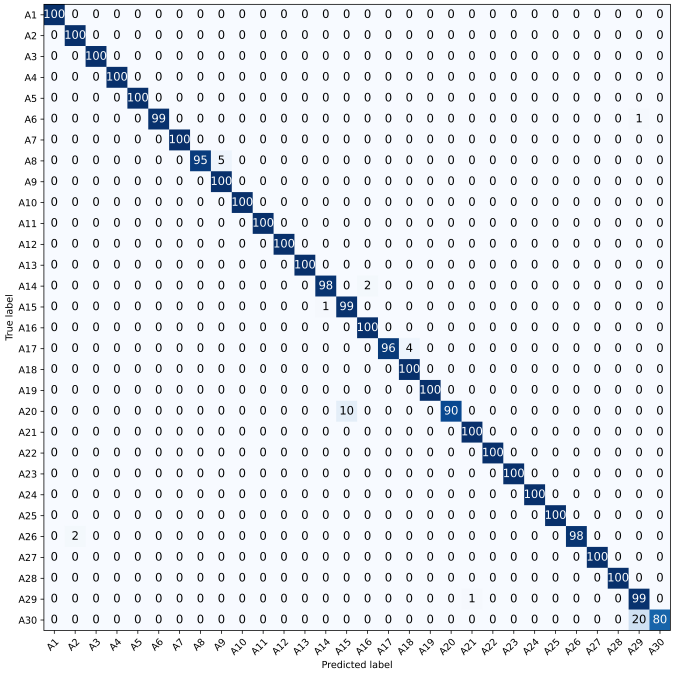


Fig. 16. Normalized confusion matrices of SOCNet. Each column of the confusion matrix represents the predicted class, each row represents the true class of the data, and the value in the grid represents the number of predicted categories of the corresponding column for the real category data of the corresponding row.

Therefore, the SOCNet model proposed in this article performs better on the task of fine-grained object recognition and is lighter and more efficient.

2) Verification on the Satellite Imitated by TX2: To prove the fast and accurate inferencing ability of SOCNet on the satellite, we set up some comparative experiments on TX2. The training and tuning of all the models are carried out on the ground server, and we inject the optimal models obtained by the ground server onto the satellite board for on-orbit object recognition [10]. The development environment configuration of TX2 is as follows: the operating system is 64-bit Ubuntu 18.04.5 LTS, the Jetpack version is Jetpack 5.1, the programming language is Python-3.6.9, and the deep learning framework is PyTorch-1.7.0.

The experimental results are shown in Figs. 13 and 14. It can be seen that SOCNet's throughput is 408.58 higher than MobiNetv3small's 330.26, ShuffleNetv2's 368.41, MnasNet's 139.11, SqueezeNet's 206.42, EMNet's 351.42, and LGRIN's 344.89. SOCNet has the lowest latency of 2.45 ms. SOCNet's accuracy is 0.99, which is 8% higher than MobiNetv3small, 9% higher than ShuffleNetv2, 7% higher than MnasNet, 12% higher than SqueezeNet, 6% higher than EMNet, and 22% higher than LGRIN. The inference time of DSHDC is comparable to MB3, SF2, EMNet, and LGRIN, but its accuracy is higher than theirs. This indicates that the large-scale feature extraction capability of DSHDC is effective. Furthermore, it can be seen from the experimental results that the object recognition accuracy, inference time, throughput, and latency of SOCNet are excellent, which shows that it is very effective for satellite on-orbit object recognition tasks. The code for SOCNet is available at https://github.com/pangyanhua/SOCNet.

V. Conclusion

This article proposes a fast and lightweight satellite on-orbit object fine-grained recognition model, SOCNet, which is designed to be lightweight from both the overall architecture and the local modules, reducing the depth, parameters' amount, size, and computation of model while taking into account multiscale, coarse-grained, and fine-grained feature extraction. The overall architecture based on the idea of flat multibranch feature extraction adopts multiscale residual parallel structure to enhance the expression of features while reducing the network depth. In the local module, to improve the large-scale feature extraction capability and feature extraction efficiency and reduce the computational cost, we innovatively propose the idea of exchanging a larger receptive field for network depth and design a DSHDC combined with the idea of DSC, which greatly reduces the number of parameters and computation. Aiming at the problem of low accuracy of object recognition methods based on RSIs due to the difficulty in extracting effective features, a feature extraction method of coupled fine–coarse-grained (CFC-module) is proposed. Finally, at the end of the network, the global average pool is used for feature fusion to further reduce the number of network parameters and computational complexity. Experiments show that the SOCNet model proposed in this article has achieved excellent performance on the task of satellite on-orbit object fine-grained recognition and is more lightweight, efficient, and faster.

In future work, first, we will explore the limits of lightweight of deep learning models using NAS with reinforcement learning and consider the comprehensive use of compression acceleration methods such as pruning, quantization, parameter sharing, and distillation to further lighten the model. Second, we will study low-power, high-efficiency, and fast new neural networks inspired by biological neural networks and develop supporting hardware computing platforms. Finally, we will apply light models to edge devices in more scenarios, such as aircraft, drones, and cameras.

References

[1] X. Wang, Y. Han, V. C. M. Leung, D. Niyato, X. Yan, and X. Chen, "Convergence of edge computing and deep learning: A comprehensive survey," *IEEE Commun. Surveys Tuts.*, vol. 22, no. 2, pp. 869–904, 2nd Quart., 2020.

[2] M. Ghiglione and V. Serra, "Opportunities and challenges of AI on satellite processing units," in *Proc. 19th ACM Int. Conf. Comput. Frontiers*, New York, NY, USA, May 2022, pp. 221–224.

[3] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *Nature*, vol. 521, no. 7553, pp. 436–444, 2015.

[4] C. Wu, H. Chen, B. Du, and L. Zhang, "Unsupervised change detection in multitemporal VHR images based on deep kernel PCA convolutional mapping network," *IEEE Trans. Cybern.*, vol. 52, no. 11, pp. 1–15, Nov. 2021.

[5] R. M. Anwer, F. S. Khan, J. van de Weijer, M. Molinier, and J. Laaksonen, "Binary patterns encoded convolutional neural networks for texture recognition and remote sensing scene classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 138, pp. 74–85, Apr. 2018.

[6] H. Ma, Y. Liu, Y. Ren, D. Wang, L. Yu, and J. Yu, "Improved CNN classification method for groups of buildings damaged by earthquake, based on high resolution remote sensing images," *Remote Sens.*, vol. 12, no. 2, p. 260, Jan. 2020.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 19:48:10 UTC from IEEE Xplore. Restrictions apply.

[7] W. Xiong, Z. Xiong, Y. Cui, and Y. Lv, "Deep multi-feature fusion network for remote sensing images," *Remote Sens. Lett.*, vol. 11, no. 6, pp. 563–571, Jun. 2020.

[8] A. Reuther, P. Michaleas, M. Jones, V. Gadepally, S. Samsi, and J. Kepner, "AI accelerator survey and trends," in *Proc. IEEE High Perform. Extreme Comput. Conf. (HPEC)*, Sep. 2021, pp. 1–9.

[9] J. Goodwill et al., "NASA spacecube edge TPU smallsat card for autonomous operations and onboard science-data analysis," in *Proc. Small Satell. Conf.*, 2021, pp. 1–13.

[10] G. Giuffrida et al., "The ϕ-Sat-1 mission: The first on-board deep neural network demonstrator for satellite earth observation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022.

[11] S. Kacker, A. Meredith, K. Cahoy, and G. Labrèche, "Machine learning image processing algorithms onboard OPS-SAT," in *Proc. 36th Annu. AIAA/USU Conf. Small Satell.*, 2022, pp. 1–13.

[12] Y. Zhou, S. Chen, Y. Wang, and W. Huan, "Review of research on lightweight convolutional neural networks," in *Proc. IEEE 5th Inf. Technol. Mechatronics Eng. Conf. (ITOEC)*, Jun. 2020, pp. 1713–1720.

[13] F. N. Iandola, S. Han, M. W. Moskewicz, K. Ashraf, W. J. Dally, and K. Keutzer, "SqueezeNet: AlexNet-level accuracy with 50X fewer parameters and <0.5 MB model size," 2016, *arXiv:1602.07360*.

[14] A. G. Howard et al., "MobileNets: Efficient convolutional neural networks for mobile vision applications," 2017, *arXiv:1704.04861*.

[15] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, "MobileNetV2: Inverted residuals and linear bottlenecks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 4510–4520.

[16] X. Zhang, X. Zhou, M. Lin, and J. Sun, "ShuffleNet: An extremely efficient convolutional neural network for mobile devices," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6848–6856.

[17] N. Ma, X. Zhang, H.-T. Zheng, and J. Sun, "ShuffleNet V2: Practical guidelines for efficient CNN architecture design," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 116–131.

[18] M. Tan and Q. V. Le, "MixConv: Mixed depthwise convolutional kernels," 2019, *arXiv:1907.09595*.

[19] M. Tan and Q. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *Proc. Int. Conf. Mach. Learn.*, 2019, pp. 6105–6114.

[20] A. Howard et al., "Searching for MobileNetV3," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2019, pp. 1314–1324.

[21] B. Zoph and Q. V. Le, "Neural architecture search with reinforcement learning," 2016, *arXiv:1611.01578*.

[22] M. Tan et al., "MnasNet: Platform-aware neural architecture search for mobile," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 2815–2823.

[23] T.-J. Yang et al., "NetAdapt: Platform-aware neural network adaptation for mobile applications," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 285–300.

[24] F. Yu and V. Koltun, "Multi-scale context aggregation by dilated convolutions," 2015, *arXiv:1511.07122*.

[25] P. Lv, W. Wu, Y. Zhong, F. Du, and L. Zhang, "SCViT: A spatial-channel feature preserving vision transformer for remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–12, 2022.

[26] A. Maskey and M. Cho, "CubeSatNet: Ultralight convolutional neural network designed for on-orbit binary image classification on a 1U CubeSat," *Eng. Appl. Artif. Intell.*, vol. 96, Nov. 2020, Art. no. 103952.

[27] S. Qiao, X.-M. Dong, J. Peng, and W. Sun, "LiteSCANet: An efficient lightweight network based on spectral and channel-wise attention for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 11655–11668, 2021.

[28] M. E. Paoletti, J. M. Haut, N. S. Pereira, J. Plaza, and A. Plaza, "Ghostnet for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 12, pp. 10378–10393, Dec. 2021.

[29] S. Jia et al., "A lightweight convolutional neural network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4150–4163, 2021.

[30] B. Cui, X.-M. Dong, Q. Zhan, J. Peng, and W. Sun, "LiteDepthwiseNet: A lightweight network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2022.

[31] J. Li, M. Weinmann, X. Sun, W. Diao, Y. Feng, and K. Fu, "Random topology and random multiscale mapping: An automated design of multiscale and lightweight neural network for remote-sensing image recognition," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–17, 2022.

[32] C. Xu, G. Zhu, and J. Shu, "A lightweight and robust lie group-convolutional neural networks joint representation for remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2022.

[33] Y. Li and C. Bian, "Few-shot fine-grained ship classification with a foreground-aware feature map reconstruction network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–12, 2022.

[34] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," *Commun. ACM*, vol. 60, no. 2, pp. 84–90, Jun. 2017.

[35] C. Szegedy et al., "Going deeper with convolutions," in *Proc. IEEE Conf. Comput. Vis. pattern Recognit.*, Jun. 2015, pp. 1–9.

[36] N. Zhang, J. Donahue, R. Girshick, and T. Darrell, "Part-based R-CNNs for fine-grained category detection," in *Proc. Eur. Conf. Comput. Vis.* Springer, 2014, pp. 834–849.

[37] S. Huang, Z. Xu, D. Tao, and Y. Zhang, "Part-stacked CNN for fine-grained visual categorization," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 1173–1182.

[38] Z. Ge, C. McCool, C. Sanderson, and P. Corke, "Subset feature learning for fine-grained category classification," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2015, pp. 46–52.

[39] Z. Wang, X. Wang, and G. Wang, "Learning fine-grained features via a CNN tree for large-scale classification," *Neurocomputing*, vol. 275, pp. 1231–1240, Jan. 2018.

[40] T. Xiao, Y. Xu, K. Yang, J. Zhang, Y. Peng, and Z. Zhang, "The application of two-level attention models in deep convolutional neural network for fine-grained image classification," in *Proc. IEEE Conf. Comput. Vis. pattern Recognit.*, Jun. 2015, pp. 842–850.

[41] B. Zhao, X. Wu, J. Feng, Q. Peng, and S. Yan, "Diversified visual attention networks for fine-grained object classification," *IEEE Trans. Multimedia*, vol. 19, no. 6, pp. 1245–1256, Jun. 2017.

[42] K.-W. Hung, Z. Zhang, and J. Jiang, "Real-time image super-resolution using recursive depthwise separable convolution network," *IEEE Access*, vol. 7, pp. 99804–99816, 2019.

[43] X. Li, Z. Du, Y. Huang, and Z. Tan, "A deep translation (GAN) based change detection network for optical and SAR remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 179, pp. 14–34, Sep. 2021.

[44] J. Hu, L. Shen, S. Albanie, G. Sun, and E. Wu, "Squeeze-and-excitation networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 8, pp. 2011–2023, Apr. 2020.

[45] L. Bai, Q. Liu, C. Li, C. Zhu, Z. Ye, and M. Xi, "A lightweight and multiscale network for remote sensing image scene classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.



Yanhua Pang (Graduate Student Member, IEEE) received the B.S. degree in automation from the Hunan University of Technology, Zhuzhou, China, in 2018, and the M.S. degree of instrumentation engineering from Yanshan University, Qinhuangdao, China, in 2021. He is currently pursuing the Dr.-Ing. degree in mechanical engineering with the Institute of Space Science and Applied Technology, Harbin Institute of Technology, Shenzhen, China.

His research interests include lightweight neural network, model compression, object detection and recognition based on remote sensing images, and satellite on-orbit computing.



Yamin Zhang received the B.Eng. degree in remote sensing science and technology from the Shandong University of Science and Technology, Qingdao, China, in 2013, and the M.A.Eng. degree in surveying and mapping engineering from the Shandong University of Science and Technology, Qingdao, China, in 2015.

He is currently a Research Assistant with the School of Harbin Institute of Technology, Shenzhen, China. His research interests include space photogrammetry and deep learning of remote sensing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 19:48:10 UTC from IEEE Xplore. Restrictions apply.



Yi Wang received the B.S. degree in remote sensing science and technology from the University of Information Engineering, Zhengzhou, China, in 2010, the M.S. degree in photogrammetry and remote sensing from the Shandong University of Science and Technology, Qingdao, China, in 2013, and the Ph.D. degree in remote sensing science and technology from Chang'an University, Xi'an, China, in 2018.

He is currently a Post-Doctoral Fellow with the School of Harbin Institute of Technology, Shenzhen, China. His research interests include space photogrammetry and deep learning of remote sensing.



Xiaofeng Wei received the Ph.D. degree in remote sensing science and technology from Information Engineering University, Zhengzhou, China, in 2015.

Then he studied as a Post-Doctoral Fellow with Peking University, Beijing, China. His research interests include remote sensing technology and global discrete grid system.



Bo Chen (Member, IEEE) received the B.E. degree in aerial photogrammetry and the Ph.D. degree in photogrammetric engineering and remote sensing from Information Engineering University, Zhengzhou, China, in 2002 and 2008, respectively.

Since 2020, he has been a full Professor with the Institute of Space Science and Applied Technology, Harbin Institute of Technology, Shenzhen, China. His research interests include satellite on-orbit computing, geospatial big data, and spatial information engineering.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 19:48:10 UTC from IEEE Xplore. Restrictions apply.